Lab 2 — Gaussian elimination · Fri Oct 23

Name: _____ **Language (circle):** Octave / Python

Quiz 7 is online in Canvas (transforms only). This ticket is the lab.

Each system has one solution. Write the elementary matrices, back-substitute by hand, then get the same x in Octave (`A \ b`) or Python (`np.linalg.solve`). Check $Ax - b$.

New syntax

I needed to…	Command / pattern
Enter A and column b	
Augmented $[A \mid b]$	
Left-multiply by E	
Solve, then check $Ax - b$	

C2.1. $x_1 + 2x_2 = 7,\ 3x_1 + x_2 = 11.$

$$E = \begin{bmatrix} __ & __ \\ __ & __ \end{bmatrix}$$

Echelon row 2: _____

$x_2 =$ _____ $x_1 =$ _____ computer $x =$ _____ $Ax-b =$ _____

C2.2. $x_1 + 3x_2 = 10,\ 2x_1 + x_2 = 5.$

$$E = \begin{bmatrix} __ & __ \\ __ & __ \end{bmatrix}$$

Echelon row 2: _____

$x_2 =$ _____ $x_1 =$ _____ computer $x =$ _____ $Ax-b =$ _____

C2.3. $x_1 + x_2 + x_3 = 6,\ x_1 + 2x_2 + x_3 = 7,\ 2x_1 + x_2 + 3x_3 = 14.$

$$E_1 = \begin{bmatrix} __ & __ & __ \\ __ & __ & __ \\ __ & __ & __ \end{bmatrix} \quad E_2 = \begin{bmatrix} __ & __ & __ \\ __ & __ & __ \\ __ & __ & __ \end{bmatrix} \quad E_3 = \begin{bmatrix} __ & __ & __ \\ __ & __ & __ \\ __ & __ & __ \end{bmatrix}$$

Echelon form: _____

_____

$x_3 =$ _____ $x_2 =$ _____ $x_1 =$ _____ computer $x =$ _____ $Ax - b =$ _____

Stuck on: _____

Check-in initials: _____